

INSTRUCTOR ANSWER KEY — DO NOT DISTRIBUTE TO STUDENTS

ECON 308 — International Trade

Lecture 1 Quiz | Fall 2026

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Quiz summary. Source: Lec1.tex (Lec1.pdf), Slides/Lec1/, Fall 2026. Total: 20 marks. Time allowed: 20 minutes.

Topics assessed: GDP expenditure identity and net exports (Q1, Part B(d)); gravity model equation and proportionality (Q2, Part B); McCallum border effect (Q3); Dutch Disease mechanism (Q5); trade-macro transmission (Part B(d)).

Part A: Multiple Choice — Answer Key

Question	Correct Answer	Marks
1	B	2
2	A	2
3	C	2
4	B	2
5	D	2
Part A Total		10

Question 1. Correct Answer: B

B) Trade *deficit* of \$40 billion; this reduces GDP by \$40 billion.

Explanation. Net exports are $NX = X - M = 800 - 840 = -\40 billion. Since $Y = C + I + G + NX$, a \$40 billion fall in NX directly reduces Y by \$40 billion (all else constant). This is a trade *deficit*, not a surplus (eliminating A and D). Option C is incorrect because exports and imports are not subtracted separately; only the *net* term $X - M$ enters the identity.

Slide reference: “The Trade-Macro Nexus” (slide 2).

Question 2. Correct Answer: A

A) Increase by exactly 20%.

Explanation. With $\alpha = 1$, the gravity model is $T_{ij} = Y_i \cdot Y_j / D_{ij}$. If Y_i increases to $1.2 Y_i$ (all else constant):

$$T'_{ij} = \frac{(1.2 Y_i) \cdot Y_j}{D_{ij}} = 1.2 \cdot \frac{Y_i \cdot Y_j}{D_{ij}} = 1.2 T_{ij}.$$

This is exactly a 20% increase. Option B would require both Y_i and Y_j to grow, compounding the effect. Options C and D are economically implausible and inconsistent with $\alpha = 1$.

Slide reference: “Reading Gravity Coefficients” (slide 10).

Question 3. Correct Answer: C

C) National borders impose substantial non-tariff costs on trade, even under free trade agreements.

Explanation. McCallum’s finding is known as the “border effect” or “home bias puzzle.” CUSFTA had already eliminated most tariffs, yet interprovincial trade was approximately 20 times larger than Canada–U.S. trade at the same distance. This demonstrates that tariffs are only part of the story: currency uncertainty, different regulatory standards, separate legal systems, and customs formalities all impose trade costs that persist even under FTAs. Anderson and van Wincoop (2003) estimated the Canada–U.S. border acts as if it adds $\approx 75,000$ km of extra distance.

Slide reference: “The Home Bias Puzzle” (slide 16); “Table 2-1” (slide 18).

Question 4. Correct Answer: B

B) World trade expanded from 1870 to 1914, then collapsed between 1914 and 1945 due to war and protectionism, before recovering and expanding again after World War II.

Explanation. Lecture 1 presents trade as unfolding in distinct waves. The **First Wave** (1870–1914) was driven by falling transport costs (steamships, railways) and the gold standard, pushing world trade to roughly 8% of global GDP. The **interwar collapse** (1914–1945) was caused by World War I, the Great Depression, and protectionist policies such as the U.S. Smoot-Hawley tariff (1930), which triggered widespread retaliation and drove world trade down by about two-thirds. The **Second Wave** (post-1945) and **Third Wave** (post-1990) saw trade recover and surpass pre-war levels, driven by GATT/WTO liberalization, containerization, and the rise of global value chains.

Why the distractors are wrong:

- **A)** Incorrect. Growth was *not* continuous — the 1914–1945 period saw a sharp collapse, not further expansion.
- **C)** Incorrect. Trade was already substantial before 1945; containerization (invented 1956) accelerated trade but was not its origin.
- **D)** Incorrect. Trade did *not* recover immediately after World War I; the interwar period (1919–1945) saw sustained decline due to the Great Depression and protectionism.

Slide reference: “Three Waves of Globalization” (slides 19–22).

Question 5. Correct Answer: D

D) A commodity boom raises foreign demand for the Canadian dollar, causing the CAD to appreciate, which makes Canadian manufactured exports more expensive for foreign buyers and reduces manufacturing competitiveness.

Explanation. Dutch Disease operates through the exchange rate channel, not directly through factor or capital markets. The mechanism is: oil/commodity boom → increased foreign demand for CAD (foreigners buy CAD to pay for Canadian oil) → CAD appreciates → price of Canadian manufactured goods in foreign currency rises → manufactured exports become less competitive → manufacturing output and employment fall. During 2007–2012, CAD reached near-parity with USD; Ontario auto and BC lumber exports fell sharply. Options A and B describe plausible economic channels but are not the Dutch Disease mechanism as presented in Lecture 1.

Slide reference: “Dutch Disease: When Resource Wealth Hurts Manufacturing” (slide 31).

Part B: Numerical Problem — Full Solution

Setup and Given Data

Canada: $Y_C = \$2,400\text{B}$. Gravity model: $T_{ij} = A \cdot Y_i \cdot Y_j / D_{ij}$, $A = 1$.

Partner	GDP (Y_j , \$B)	Distance (D , km)
Australia	1,600	12,000
Mexico	1,200	3,000
United States	25,000	1,000

(a) Compute T_{CA} , T_{CM} , T_{CU} . (2 points)

Solution

Canada–Australia:

$$T_{CA} = \frac{2,400 \times 1,600}{12,000} = \frac{3,840,000}{12,000} = \mathbf{320} \text{ (in \$billion-km units)}$$

Canada–Mexico:

$$T_{CM} = \frac{2,400 \times 1,200}{3,000} = \frac{2,880,000}{3,000} = \mathbf{960}$$

Canada–United States:

$$T_{CU} = \frac{2,400 \times 25,000}{1,000} = \frac{60,000,000}{1,000} = \mathbf{60,000}$$

Grading Rubric

2 pts total. Award **1 pt** for correct formulas applied to all three pairs (partial credit for 2/3 correct). Award **1 pt** for correct arithmetic throughout.

(b) Ratio T_{CU}/T_{CA} and decomposition. (2 points)

Solution**Ratio:**

$$\frac{T_{CU}}{T_{CA}} = \frac{60,000}{320} = \mathbf{187.5}$$

Decomposition into gravity factors:

$$\frac{T_{CU}}{T_{CA}} = \underbrace{\frac{Y_{US}}{Y_{Aus}}}_{\text{Size factor}} \times \underbrace{\frac{D_{CA}}{D_{CU}}}_{\text{Proximity factor}} = \frac{25,000}{1,600} \times \frac{12,000}{1,000} = 15.625 \times 12 = \mathbf{187.5\checkmark}$$

Interpretation:

- **Size factor (15.6×):** The U.S. economy is 15.6 times larger than Australia's. Both produce more goods to sell and can buy more imports, proportionally expanding bilateral trade.
- **Proximity factor (12×):** Australia is 12 times farther from Canada than the U.S. Distance raises transport costs, information barriers, and time-in-transit costs, suppressing trade with Australia.
- **Combined:** The two factors multiply: Canada is predicted to trade **187.5 times more** with the U.S. than with Australia.

Grading Rubric

2 pts total. 0.5 pt for correct ratio (187.5). **0.5 pt** for correct algebraic decomposition. **1 pt** for clear economic interpretation of both factors (size and distance). Accept equivalent verbal explanations that correctly identify both factors.

(c) **CPTPP effect on Canada–Australia trade.** (2 points)

Solution

New effective distance: $D_{CA}^{CPTPP} = 12,000 \times 0.75 = 9,000$ km.

New predicted trade:

$$T_{CA}^{CPTPP} = \frac{2,400 \times 1,600}{9,000} = \frac{3,840,000}{9,000} \approx \mathbf{426.7}$$

Percentage increase:

$$\% \Delta = \frac{T_{CA}^{CPTPP} - T_{CA}}{T_{CA}} \times 100 = \frac{426.7 - 320}{320} \times 100 = \frac{106.7}{320} \times 100 \approx \mathbf{33.3\%}$$

Alternative derivation: A 25% reduction in distance multiplies T by $D_{old}/D_{new} = 12,000/9,000 = 4/3$, so trade rises by $4/3 - 1 = 1/3 \approx 33.3\%$.

Economic interpretation: CPTPP eliminates tariffs and reduces regulatory barriers, cutting effective distance to Australia from 12,000 to 9,000 km. This raises predicted Canada–Australia trade by one-third. Note, however, that even after CPTPP, $T_{CU}/T_{CA}^{CPTPP} = 60,000/426.7 \approx 141$: the US still dominates Canadian trade overwhelmingly, consistent with the gravity model’s prediction that proximity is difficult to overcome with policy alone.

Grading Rubric

2 pts total. 1 pt for correct $T_{CA}^{CPTPP} \approx 426.7$. **1 pt** for correct percentage increase (33.3%; accept 33% or 1/3). Deduct 0.5 pt if student calculates a percentage decrease or uses 12,000 km incorrectly. Bonus (no extra points): award full credit to students who note the policy implication about US trade dominance persisting.

(d) **US recession shock to Canadian GDP.** (4 points)

Solution

Initial conditions: $Y = \$2,400\text{B}$; $NX = +\$48\text{B}$; $C+I+G = Y - NX = 2,400 - 48 = \$2,352\text{B}$.

Shock: $\Delta X = -\$96\text{B}$; $\Delta M = 0 \Rightarrow \Delta NX = -\96B .

(i) **New net exports:**

$$NX_{\text{new}} = +48 + (-96) = \mathbf{-\$48 \text{ billion}}$$

(trade balance turns from surplus to deficit)

(ii) **New GDP:**

$$Y_{\text{new}} = (C + I + G) + NX_{\text{new}} = 2,352 + (-48) = \mathbf{\$2,304 \text{ billion}}$$

Equivalently: $\Delta Y = \Delta NX = -\96B , so $Y_{\text{new}} = 2,400 - 96 = \$2,304\text{B}$. ✓

Summary: The \$96B export fall causes a **\$96B direct reduction in Canadian GDP** (a 4% decline), shifting the trade balance from a \$48B surplus to a \$48B deficit.

Grading Rubric

4 pts total. 2 pts for (i): correct new $NX = -\$48B$ (sign required). **2 pts** for (ii): correct new $GDP = \$2,304B$. Accept either of the two approaches shown. Common errors: (a) forgetting that $\Delta M = 0$ so $\Delta NX = \Delta X$; (b) confusing the sign convention for deficit vs. surplus.

Point Summary

Part	Question / Sub-part	Marks	Cumulative
A	Q1 — GDP identity and trade balance	2	2
A	Q2 — Gravity model proportionality	2	4
A	Q3 — McCallum border effect	2	6
A	Q4 — Globalization waves (1870–present)	2	8
A	Q5 — Dutch Disease mechanism	2	10
B	(a) Gravity calculations (3 pairs)	2	12
B	(b) Ratio decomposition + interpretation	2	14
B	(c) CPTPP effect + percentage change	2	16
B	(d) Export shock to NX and GDP	4	20
Total		20	

Suggested grade boundaries (20-mark quiz, 20 minutes):

A range: 18–20 B range: 15–17 C range: 12–14 D: 10–11 F: <10